

Charles Zhao

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WORK EXPERIENCE

Lawrence Livermore National Laboratory

Summer 2023 & 2024

Cybersecurity Intern, 40 hrs/wk

Livermore, CA

- Wrote Python unit and integration tests using pytest, sqlite3, and psycpg2, reducing debugging effort and validating application compatibility with the CI/CD pipeline.
- Built a dynamic web application for the Department of Homeland Security using the MEAN stack (MongoDB, Express.js, Angular, Node.js), delivering a scalable and efficient solution for enhanced data management.
- Developed a Python script to automate report parsing and identify suspicious IP addresses and domains interacting with the network, improving threat detection and security monitoring.

PROJECTS

Thematic Investment Tool

Apr 2026 – Present

- Developing a web application that helps investors track thematic and sector performance, identify trends, assess market sentiment and the macro environment, monitor market breadth, and analyze unusual options flow and positioning data.

KubeRange

Mar – Apr 2026

- Developed a full-stack web application (React, FastAPI, PostgreSQL) hosted on Kubernetes that dynamically provisions and tears down isolated vulnerable web app instances (DVWA, Juice Shop) on demand, with an integrated ELK stack for data collection and observability.

Cascading Detection Engine

Feb – Apr 2026

- Built a multi-stage phishing detection engine combining rule-based analysis, SVM, Isolation Forest, and DistilBERT, escalating ambiguous samples between stages and using logistic regression for final score fusion.

Cybersecurity Daily Digest

Feb – Apr 2026

- Built an automated cybersecurity news pipeline using RSS feeds, GitHub Actions, and an LLM to collect, summarize, and deliver a daily digest.

Domain Sinkhole & Network Traffic Analysis

Nov – Dec 2025

- Deployed Pi-hole on a Raspberry Pi as a local DNS sinkhole to filter and block unwanted domains across all devices on a home network. Investigated captured DNS queries using Wireshark and documented findings on GitHub.

Custom C++ Stager

Aug 2024 – May 2025

- Developed a custom C++ stager within a sandboxed Windows environment to study and evaluate modern antivirus evasion techniques, including process injection and dynamic API resolution.

Malicious File/URL Scanner

Jan – May 2024

- Developed a Python microservice hosted on Proxmox that scanned email attachments and URLs using the VirusTotal API and database to identify malicious content.
- Validated detection capabilities in an isolated virtual environment using known malicious files and URLs from public sources.

EDUCATION

Virginia Tech

- **Master of Engineering, Computer Engineering** – May 2025 | GPA: 4.0
 - **Bachelor of Science, Computer Engineering, Magna Cum Laude** – May 2024 | GPA: 3.61
- Programs: SFS CyberCorps Scholar; IC CAE

CERTIFICATIONS & SKILLS

- **Certifications:** CompTIA Security+; Cranium AI Red Team Certification
- **Skills:** Python; C++; Linux; Kubernetes; Proxmox; Wireshark; Burp Suite; Testing & Automation; Ghidra